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MOLDING METHOD

Abstract

Disclosed herein is a molding method capable of reducing contamination of a base material. A molding method for molding a seal member onto a base material includes: a preforming step of injecting a rubber material into a groove at a temperature where the rubber material is not crosslinked, in a cavity plate including the groove into which the rubber material is injected; a coating step of coating a surface of the rubber material with an adhesive or pressure-sensitive adhesive, the rubber material being injected into the groove; and a vulcanization molding step of sandwiching the base material between the cavity plate on which the surface of the rubber material is coated with the adhesive or pressure-sensitive adhesive and a second mold, and molding the rubber material onto the base material as the seal member via the adhesive or pressure-sensitive adhesive at a temperature where the rubber material is crosslinked.

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Background/Summary

This application is a National Phase entry of International Application No. PCT/JP2021/026896 under § 371, filed Jul. 16, 2021, which is hereby incorporated by reference in its entirety.

FIELD OF THE DISCLOSURE

[0001] The present disclosure relates to a molding method for molding a seal member onto a base material.

BACKGROUND

[0002] In the related art, a molding method for molding a seal member such as rubber onto a base material such as metal, resin, or paper has been proposed. FIG. 6 is a cross-sectional view of a molding machine 1000 that performs transfer molding of the related art, FIG. 6(a) is a view showing a state where a rubber material 1050 is charged into a pot 1022 of the molding machine 1000, and FIG. 6(b) is a view showing a state where mold clamping and the injection of the rubber material 1050 are performed in the molding machine 1000. The molding machine 1000 includes heating plates 1010 and 1012 and molds 1020 and 1032. The mold 1020 includes the pot 1022 into which the rubber material 1050 is charged; a groove 1026 for molding a seal member onto a base material 1040; and a gate 1024 for injecting the rubber material 1050 into the groove 1026. The mold 1032 includes a recess 1033 in which the base material 1040 is disposed. The molding machine 1000 applies, for example, a pressure (hereinafter, referred to as an injection pressure) of 1 MPa to 200 MPa to the rubber material 1050 by clamping the molds after the rubber material 1050 is charged and the base material 1040 is disposed, and the rubber material 1050 is injected into the groove 1026. The molding machine 1000 molds the seal member onto the base material 1040 such as a separator by crosslinking the rubber material 1050 using, for example, heat at 80° C. to 220° C. generated by the heating plate 1010.

[0003] In addition, for example, Patent Document 1 discloses a method for manufacturing a gasket-integrated plate used for a separator for a fuel cell, a top cover of a hard disk drive, and the like. In addition, adherence between a base material and a seal member may be improved by coating a portion of the base material, onto which the seal member is formed, with an adhesive or pressure-sensitive adhesive.

[0004] Patent Document 1: JP 2004-225721 A

SUMMARY

[0005] However, in the molding method of the related art, since the base material is coated with an adhesive or pressure-sensitive adhesive, the adhesive or pressure-sensitive adhesive with which the base material is coated may flow and peel off from the base material due to an injection pressure, so that adhesion or pressure-sensitive adhesion deteriorates, which is a risk. In addition, the adhesive or pressure-sensitive adhesive with which a surface of the base material is coated by direct coating may scatter and contaminate the base material. For this reason, it is required to reduce contamination of the base material caused by the adhesive or pressure-sensitive adhesive during molding of the seal member.

[0006] The present invention has been made in view of the above-described circumstances, and an exemplary object of the present invention is to provide a molding method capable of reducing contamination of a base material caused by an adhesive or pressure-sensitive adhesive.

[0007] In order to solve the above-described problems, the present invention has the following embodiments. [0008] (1) A molding method for molding a seal member onto a base material includes: a first step of injecting a raw material for the seal member into a groove at a temperature

where the raw material is not crosslinked, in a first mold including the groove into which the raw material is injected; a second step of coating a surface of the raw material with an adhesive or a pressure-sensitive adhesive, the raw material being injected into the groove of the first mold; and a third step of sandwiching the base material between the first mold on which the surface of the raw material is coated with the adhesive or the pressure-sensitive adhesive and a second mold, and molding the raw material onto the base material as the seal member via the adhesive or the pressure-sensitive adhesive at a temperature where the raw material is crosslinked.

[0009] Further objects or other characteristics of the present invention will become apparent by a preferred embodiment that will be described below with reference to the accompanying drawings.

[0010] According to embodiments of the present invention, it is possible to provide the molding method capable of reducing contamination of the base material caused by the adhesive or pressure-sensitive adhesive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. 1 is a cross-sectional view showing a preforming step of an embodiment, FIG. 1(a) is a view showing a state where a rubber material is charged into a pot of a forming machine, and FIG. 1(b) is a view showing a state where mold clamping and the injection of the rubber material are performed in the forming machine;

[0012] FIG. 2 is a cross-sectional view showing main parts of a molding machine in a vulcanization molding step of the embodiment, FIG. 2(a) is a view showing a state where one pair of a cavity plate and a base material are disposed in the molding machine, FIG. 2(b) is a view showing a state where a plurality of pairs of the cavity plates and the base materials are disposed in the molding machine, FIG. 2(c) is a cross-sectional view showing the base material after molding, and FIG. 2(d) is a view showing a state where an inner region of the base material is sealed by a plate;

[0013] FIG. 3 is a view showing a configuration of the cavity plate of the embodiment, FIG. 3(a) is a view showing a surface of the cavity plate on which a groove is provided, FIG. 3(b) is a cross-sectional view taken along line A-A in FIG. 3(a), and FIG. 3(c) is a perspective view of a cross-section along line B-B in FIG. 3(a);

[0014] FIG. 4(a) is a cross-sectional view of main parts showing the cavity plate after the preforming step of the embodiment, FIG. 4(b) is a cross-sectional view of main parts showing the cavity plate in which the rubber material injected into the groove is coated with an adhesive or pressure-sensitive adhesive in a coating step, FIG. 4(c) is a cross-sectional view of main parts showing a state where the base material is sandwiched between the cavity plate and a mold in the vulcanization molding step, and FIG. 4(d) is a cross-sectional view of main parts showing the base material on which a seal member is molded after the vulcanization molding step;

[0015] FIG. 5(a) is a top view of main parts showing a configuration of a coating device in the coating step of the embodiment, and FIG. 5(b) is a front view of main parts showing a coating operation by the coating device; and

[0016] FIG. 6 is a cross-sectional view of a molding machine in an example of the related art, FIG. 6(a) is a view showing a state where a rubber material is charged into a pot of the molding machine, and FIG. 6(b) is a view showing a state where mold clamping and the injection of the rubber material are performed in the molding machine.

DETAILED DESCRIPTION

[0017] Hereinafter, embodiments will be described with reference to the drawings. Here, the term “forming” refers to a step of shaping rubber in an uncrosslinked state regardless of the use of molds, and the term “molding” refers to a step of shaping rubber (crosslinked rubber) by using

molds and performing a crosslinking reaction.

Embodiment

<Flow Until Base Material With Seal Member is Completed>

[0018] A basic flow of a molding method of the present embodiment will be described. When the seal member is, for example, rubber, materials for a rubber material are mixed and kneaded. Examples of the rubber used include thermosetting elastic bodies such as fluororubber, EPDM, NBR, CR, silicone, thermoplastic elastic bodies, thermoplastic resins, and the like. Next, vulcanization promoter mixing is performed on the kneaded materials. Here, vulcanization promoter mixing refers to adding, mixing, and kneading a vulcanization accelerator and a vulcanizing agent with a rubber compound. Incidentally, the material in a state where rubber used as a raw material and a compounding agent such as a filler and a crosslinking agent (vulcanizing agent) are uniformly mixed is also referred to as a rubber compound (rubber material). Next, the rubber material on which vulcanization promoter mixing is performed is dispensed and formed, and then the rubber material is injected into a groove of a cavity plate to be described later at a predetermined pressure, and is formed in an unvulcanized (uncrosslinked) state. In the following description, a step of performing forming in an unvulcanized (uncrosslinked) state is referred to as a preforming step.

[0019] In the preforming step, vulcanization (crosslinking) is performed in a state where the cavity plate into which the rubber is injected and a base material are laminated to form a pair. In the following description, a step of performing vulcanization (crosslinking) is referred to as a vulcanization molding step. Incidentally, the vulcanization molding step also includes a case where a substance used as a raw material for the seal member is molded onto the base material at a vulcanization (crosslinking) temperature. After the vulcanization (crosslinking) is completed, and for example, secondary vulcanization is performed on the base material onto which the seal member is transferred (molded), finally, finishing, inspection, and the like are performed, a series of work is completed, and the base material with the seal member is completed as a product.

(Preforming Step)

[0020] The preforming step that is a first step will be described. FIG. 1 is a cross-sectional view showing a forming machine **100** in the preforming step of the present embodiment, FIG. 1(a) is a view showing a state where a rubber material **150** that is a raw material for the seal member formed onto the base material is charged into a pot **122** of the forming machine **100**, and FIG. 1(b) is a view showing a state where the rubber material **150** is injected into a groove **162** of a cavity plate **160** in the forming machine **100**. In FIG. 1, an up-down direction is indicated by a double-headed arrow. Incidentally, in FIGS. 1 to 3, for the sake of description, it is assumed that the base material is a flat plate and the cavity plate **160** of which the shape is determined according to the base material is a flat plate.

[0021] The forming machine **100** includes heating plates **110** and **112** and molds **120** and **130**. The mold **120** includes the pot **122** into which the rubber material **150** is charged, and a gate **124** for injecting the rubber material **150** into the groove **162** of the cavity plate **160**. The mold **130** includes a recess **133** in which the cavity plate **160** is disposed.

[0022] The cavity plate **160** that is a first mold includes the groove **162** into which the rubber material **150** is injected, and a groove (hereinafter, referred to as an escape groove) **164** for the escape of the rubber material **150** from the groove **162** in the preforming step or the vulcanization molding step. In addition, the cavity plate **160** also includes connecting grooves (not shown) to be described later.

[0023] In the forming machine **100**, as shown in FIG. 1(a), the rubber material **150** is charged into the pot **122**, and the cavity plate **160** is disposed in the recess **133** of the mold **130**. Incidentally, it is assumed that positioning between the mold **120** (gate **124**) and the mold **130** (the groove **162** of the cavity plate **160**) is performed by a known method or the like. Thereafter, as shown in FIG. 1(b), a pressure (hereinafter, also referred to as a mold clamping pressure) is applied to the molds

120 and **130** to clamp the molds, and a pressure for injecting the rubber material **150** from the pot **122** via the gate **124** is applied to inject the rubber material **150** into the groove **162** of the cavity plate **160**. The pressure required when injecting the rubber material **150** is referred to as an injection pressure, and is, for example, 1 to 200 MPa. In addition, the temperature of heat generated in the heating plates **110** and **112** during mold clamping shown in FIG. 1(b) is a temperature where the rubber material **150** is not vulcanized (crosslinked) (in other words, unvulcanized (uncrosslinked)) (the substance used as a raw material for the seal member is not crosslinked) and fluidity is maintained. For example, in the present embodiment, the preforming step is performed at a temperature of 80° C. In this respect, the preforming step can also be referred to as an unvulcanized forming step. Incidentally, the temperature in the preforming step is not limited to 80° C., and is set depending on, for example, the viscosity of the rubber material **150**. The filling rate of the rubber material **150** into the groove **162** of the cavity plate **160** is, for example, 90% to 110%; however, the filling rate is also not limited to this value.

(Vulcanization Molding Step)

[0024] The vulcanization molding step that is a third step will be described. FIG. 2 is a cross-sectional view showing main parts of a molding machine **200** in the vulcanization molding step of the present embodiment, FIG. 2(a) is a view showing a state where one pair of the cavity plate **160** and a base material **140** are disposed in the molding machine **200**, FIG. 2(b) is a view showing a state where a plurality of pairs (for example, three pairs) of the cavity plates **160** and the base materials **140** are disposed in the molding machine **200**, FIG. 2(c) is a cross-sectional view showing the base material **140** after molding, and FIG. 2(d) is a view showing a state where an inner region of the base material **140** is sealed by a plate **180**. In FIG. 2, the up-down direction is indicated by a double-headed arrow. Incidentally, since a second pair and a third pair have the same configuration as a first pair in FIG. 2(b), reference signs are omitted.

[0025] The molding machine **200** includes heating plates **210** and **212** and a second mold **222** (hereinafter, simply referred to as the mold **222**). The base material **140** is, for example, metal, resin, or paper. The base material **140** has a surface **142** onto which a seal member **172** is formed, and a surface **144** opposite to the surface **142**. The cavity plate **160** in which the rubber material **150** is injected into the groove **162** in the preforming step is aligned with the surface **142** of the base material **140**, and the mold **222** is aligned with the surface **144**. Namely, the base material **140** is sandwiched between the cavity plate **160** as a mold and the mold **222**. Incidentally, it is assumed that positioning between the cavity plate **160** (the groove **162** filled with the rubber material **150**) and the base material **140** is performed by a known method or the like.

[0026] In this state, heating is performed by the heating plates **210** and **212** at a temperature where the rubber material **150** is vulcanized (crosslinked) (a substance used as a raw material for the seal member is crosslinked), for example, at 120° C. to 220° C., and a predetermined pressure (for example, 5 MPa) is applied in the up-down direction. Here, since the injection of the rubber material **150** into the groove **162** is completed in the preforming step, the predetermined pressure does not include the injection pressure but only the mold clamping pressure. In other words, the injection pressure in the vulcanization molding step is 0 MPa. In addition, when the predetermined pressure and heat are applied, the rubber material **150** in the groove **162** escapes into the escape groove **164** via the connecting grooves (not shown) to be described later. Incidentally, the rubber material **150** in the groove **162** may escape to the escape groove **164** during the preforming step.

[0027] Incidentally, FIG. 2(a) shows an example in which the vulcanization molding step is performed on one pair of the cavity plate **160** and the base material **140**; however, embodiments of the present invention are not limited thereto. For example, as shown in FIG. 2(b), molding may be performed on a plurality of pairs, for example, three pairs of the cavity plates **160** and the base materials **140** in one vulcanization molding step. In addition, in FIG. 2, the cavity plate **160** is at the top and the base material **140** is at the bottom; however, the top and bottom may be reversed. When the top and bottom are reversed, the mold **222** becomes an upper mold.

[0028] As described above, as shown in FIG. 2(c), the rubber material **150** injected into the groove **162** of the cavity plate **160** is transferred (molded) onto the surface **142** of the base material **140** as the seal member **172**. In addition, as shown in FIG. 2(d), by aligning the plate **180** to face the surface **142** of the base material **140** on which the seal member **172** is molded, the inner region is sealed by the seal member **172**. For example, gas, liquid, or the like is retained in the sealed region. Incidentally, hereinafter, a vulcanized (crosslinked) portion of the rubber material **150** that has escaped to the escape groove **164** is referred to as a side lip **174**. Here, in FIGS. 1 and 2, the seal member **172** is provided on the surface **142** of the base material **140**; however, the seal member **172** may be provided on the surface **144** of the base material **140** or both surfaces (the surface **142** and the surface **144**) of the base material **140**.

(Cavity Plate)

[0029] A configuration of the cavity plate **160** will be described. FIG. 3 is a view showing a configuration of the cavity plate **160**, FIG. 3(a) is a view showing a surface (hereinafter, referred to as a touch surface) **168** of the cavity plate **160** on which the groove **162** is provided, FIG. 3(b) is a cross-sectional view taken along line A-A in FIG. 3(a), and FIG. 3(c) is a perspective view of a cross-section along line B-B in FIG. 3(a). The cavity plate **160** includes the groove **162**, the escape groove **164**, and connecting grooves **166**.

[0030] As described above, the groove **162** is a groove into which the rubber material **150** is injected. For example, in the present embodiment, it is assumed that the base material **140** has a rectangular shape having two long sides and two short sides and the seal member is molded into a rectangular frame shape along end portions of the base material **140**. For this reason, as shown in FIG. 3(a), similarly, the groove **162** of the cavity plate **160** is also provided in a rectangular frame shape. The groove **162** has a predetermined depth D1 (refer to FIG. 3(c)). The depth D1 of the groove **162** determines a height of the seal member molded onto the base material **140**.

Incidentally, the molding position, length, and width of the seal member on the base material **140** are determined depending on the purpose of use of the base material **140** and/or the seal member, and are not limited to a shape and the like shown in FIG. 3.

[0031] The escape groove **164** is provided parallel to and spaced apart from the groove **162**, but may not be parallel thereto. The escape groove **164** is a groove for the escape of surplus rubber when the rubber material **150** injected into the groove **162** in the vulcanization molding step (or the preforming step) expands during heating, exceeds the volume of the groove **162**, and becomes surplus. Namely, the occurrence of burrs after the vulcanization molding step can be suppressed by providing the escape groove **164**. As shown in FIG. 3(a), for example, in the present embodiment, the escape groove **164** is provided between the groove **162** and four sides (end portions) of the cavity plate **160**. In the present embodiment, the escape groove **164** is continuously provided, and similarly to the groove **162**, has a rectangular frame shape.

[0032] Incidentally, the escape groove **164** may be provided discontinuously. In addition, the escape groove **164** is provided at a position where the purpose of use of the base material **140** on which the seal member **172** is molded is not hindered. For example, in the case of the base material **140** onto which the seal member **172** is molded using the cavity plate **160** in FIG. 3(a), the seal member **172** is molded to retain sealability of a region inside the seal member **172** molded by the groove **162**. For this reason, in the cavity plate **160**, the escape groove **164** is provided at a position corresponding to a region outside the seal member **172** (namely, the end portions of the base material **140**), which does not require the retention of sealability.

[0033] The escape groove **164** has a predetermined depth D2 (refer to FIG. 3(c)). The depth D2 of the escape groove **164** determines a height of the side lip **174**. Here, the depth D2 of the escape groove **164** is shallower than the depth D1 of the groove **162** ($D2 < D1$). In other words, the height of the side lip **174** is lower than the height of the seal member **172**. This is intended to retain sealability of the region inside the seal member **172** when the plate **180** is aligned with the base material **140** by making the height of the seal member **172** higher than the height of the side lip **174**.

(refer to FIG. 2(d)). The depth D2 of the escape groove **164**, whether the escape groove **164** is continuous or discontinuous, or the length, width, shape, and the like when the escape groove **164** is discontinuous are not limited to those shown in FIG. 3, and may be set depending on the purpose of use of the base material **140** and/or the seal member **172** or the viscosity, injection amount, injection pressure, and the like of the rubber material **150**.

[0034] The connecting grooves **166** are grooves through which the rubber material **150** expanded by heat in the vulcanization molding step (or the preforming step) escapes into the escape groove **164**. The connecting grooves **166** have, for example, a depth d of approximately 0.005 mm to approximately 0.2 mm and a length L of approximately 1 to 6 mm, and are formed to be lower than the touch surface **168**. Here, the length L is a length in a direction parallel to the groove **162**. Incidentally, the depth d of the connecting grooves **166** is shallower than the depth D1 of the groove **162** and the depth D2 of the escape groove **164** ($d < D2 < D1$).

[0035] As shown in FIG. 3(a), the connecting grooves **166** are discretely provided between the groove **162** and the escape groove **164**. The positions where the connecting grooves **166** are provided, the number, width (in other words, a distance between the groove **162** and the escape groove **164**), the length L, and the depth d of the connecting grooves **166** may be set depending on the purpose of use of the base material **140** and/or the seal member **172**, the configurations of the groove **162** and the escape groove **164**, the viscosity of the rubber material **150**, the position of the gate **124**, or the like.

[0036] For example, metals such as iron, SUS, aluminum, and copper that can withstand a temperature during vulcanization (hereinafter, referred to as a vulcanization (crosslinking) temperature) in the vulcanization molding step and that have good thermal conductivity are suitable for the material of the cavity plate **160**. In addition, as the material of the cavity plate **160**, for example, ceramic, resin, or the like can also be used as long as these materials satisfy the above-described conditions. In addition, it is assumed that the cavity plate **160** has such a thickness that deformation is prevented and rigidity is maintained in the preforming step and the vulcanization molding step.

[0037] Since the volumes of the groove **162** and the escape groove **164** of the cavity plate **160** are known values at the time of designing the cavity plate **160**, the amount of surplus rubber can be controlled by the escape groove **164**, and the occurrence of burrs can be reduced. In addition, since the injection of the rubber material **150** is performed in the preforming step, no injection pressure is applied when the rubber material **150** is molded onto the base material **140** in the vulcanization molding step. For this reason, deformation or damage of the base material **140** caused by the injection pressure can be reduced. In addition, since the injection pressure is not applied in the vulcanization molding step, the pressure added to the mold clamping pressure to suppress the occurrence of burrs in the related art can be reduced. Further, for example, gate marks remain in transfer molding, injection molding, or the like; however, in the preforming step of the present embodiment, since the rubber material **150** is injected into the groove **162** of the cavity plate **160** in an unvulcanized state, gate marks can be reduced or eliminated.

<Coating Step>

[0038] The molding method including a coating step that is a second step of the present embodiment will be described using FIG. 4. FIG. 4(a) is a cross-sectional view of main parts showing the cavity plate **160** after the preforming step described in FIG. 1, and FIG. 4(b) is a cross-sectional view of main parts showing the cavity plate **160** in which the rubber material **150** injected into the groove **162** is coated with an adhesive or pressure-sensitive adhesive **600** in the coating step of the present embodiment. In addition, FIG. 4(c) is a cross-sectional view of main parts showing a state where the base material **140** is sandwiched between the cavity plate **160** and the mold **222** in the vulcanization molding step described in FIG. 2, and FIG. 4(d) is a cross-sectional view of main parts showing the base material **140** on which the seal member **172** is molded after the vulcanization molding step. Incidentally, in FIG. 4, for the sake of description, the adhesive or

pressure-sensitive adhesive **600** is depicted to have a thickness; however, actually, a thickness of the adhesive or pressure-sensitive adhesive **600** is, for example, 0.01 to 200 μm . In addition, the cavity plate **160**, the base material **140**, and the like have the same configurations as the configurations described in FIGS. **1** to **3**, and reference signs and lead lines for some portions of FIG. **4** are omitted. Further, since the adhesive or pressure-sensitive adhesive **600** is shown in black, the rubber material **150**, the seal member **172**, and the side lip **174** are shown with grid-pattern hatching.

[0039] As shown in FIG. **4(a)**, the groove **162** of the cavity plate **160** is filled (injected) with the rubber material **150** by injection. Incidentally, the rubber material **150** does not escape into the escape groove **164** in FIG. **4(a)**; however, during the preforming step, the rubber material **150** may escape thereinto via the connecting grooves **166** (refer to FIG. **3**).

[0040] In the present embodiment, when the preforming step is completed, the process proceeds to the coating step. In the coating step, for example, the coating of the adhesive or pressure-sensitive adhesive **600** is performed by a coating device (not shown). In the present embodiment, the coating of the adhesive or pressure-sensitive adhesive **600** by the coating device is not performed on the base material **140** but on the cavity plate **160**. Specifically, the coating device coats the portion of the rubber material **150** injected into the groove **162** of the cavity plate **160**, namely, a surface of the rubber material **150** injected into the groove **162** (an exposed portion of the rubber material **150** in the groove **162**) with the adhesive or pressure-sensitive adhesive **600**.

[0041] Incidentally, in the coating step, the coating of the adhesive or pressure-sensitive adhesive **600** may be performed by replacing components above the mold **120** with the coating device in a state where the cavity plate **160** is disposed in the forming machine **100** after the preforming step. In addition, before the vulcanization molding step, in a state where the cavity plate **160** is disposed in the molding machine **200**, the coating of the adhesive or pressure-sensitive adhesive **600** may be performed by the coating device before the base material **140** is superimposed on the cavity plate **160**. Further, between the preforming step and the vulcanization molding step, the coating of the adhesive or pressure-sensitive adhesive **600** may be performed in a state where the cavity plate **160** is removed from the forming machine **100** and is disposed in the coating device. As shown in FIG. **4(b)**, the rubber material **150** injected into the groove **162** of the cavity plate **160** after the preforming step may be coated with the adhesive or pressure-sensitive adhesive **600**, and the method is not limited to the above-described methods.

[0042] In FIG. **4(c)**, the vulcanization molding step is performed in a state where the rubber material **150** injected into the groove **162** of the cavity plate **160** is coated with the adhesive or pressure-sensitive adhesive **600**. As shown in FIG. **4(d)**, the seal member **172** is more firmly transferred (molded) onto the base material **140** by the adhesive or pressure-sensitive adhesive **600**.

<Coating Device for Adhesive>

[0043] A coating device that automatically coats the rubber material **150** with the adhesive or pressure-sensitive adhesive **600**, the rubber material **150** being injected into the groove **162** of the cavity plate **160**, will be described. The coating device is, for example, a spray gun, a dispenser, a coater, a printer, or the like. Incidentally, when there is a risk that the coating device coats an unnecessary region with the adhesive or pressure-sensitive adhesive **600**, for example, a method such as masking may be used to prevent the unnecessary coating of the adhesive or pressure-sensitive adhesive **600**. Here, the printer is, for example, an inkjet printer.

[0044] Here, an example in which an inkjet printer is used as a coating device **500** will be described using FIG. **5**. FIG. **5(a)** is a top view of main parts showing a configuration of the coating device **500** in the coating step, and FIG. **5(b)** is a front view of main parts showing a coating operation by the coating device **500**. FIG. **5(b)** is shown in the up-down direction. Incidentally, in FIG. **5**, some of the reference signs described up to now are omitted. The coating device **500** includes a head **510**, a table **520**, rails **512** and **522**, and a control unit **540**. The head **510** includes a nozzle **530** on the lower side. A bottle **610** containing the adhesive or pressure-

sensitive adhesive **600** is placed in the head **510**, and supplies the adhesive or pressure-sensitive adhesive **600** to the nozzle **530** via a tube (not shown). The nozzle **530** is provided with a plurality of holes (not shown) that eject the adhesive or pressure-sensitive adhesive **600** toward the lower side supplied from the bottle **610**. The head **510** is movable along the rail **512** at a predetermined speed. A movement direction of the head **510** is, hereinafter, an X direction.

[0045] The table **520** is a stand on which the cavity plate **160** that has undergone the completion of the preforming step is placed. The cavity plate **160** that has undergone the completion of the preforming step is placed on the table **520** such that the rubber material **150** faces upward. The table **520** is movable along the rail **522** at a predetermined speed. A movement direction of the table **520** is, hereinafter, a Y direction. The cavity plate **160** is positioned on the table **520** by a known method, and is fixed by a known method such that the cavity plate **160** does not move during movement of the table **520**.

[0046] The control unit **540** includes various means for controlling the coating device **500**, such as a CPU (not shown), a ROM (not shown), a RAM (not shown), a timer (not shown), and a counter (not shown). The control unit **540** controls the position and movement speed of the head **510**, the position and movement speed of the table **520**, the ejection amount of the adhesive or pressure-sensitive adhesive **600** from the nozzle **530**, the ejection timing, and the like using a known method.

[0047] Since the position of the groove **162** of the cavity plate **160** shown in FIG. 3(a) is already known when the cavity plate **160** is designed, position information of the groove **162** is obtained in advance. For this reason, the coating operation of the adhesive or pressure-sensitive adhesive **600** is controlled by the control unit **540** included in the coating device **500**, based on the position information of the groove **162** of the cavity plate **160**. For example, when the coating device **500** is an inkjet printer, the position where the coating of the adhesive or pressure-sensitive adhesive **600** is performed can be controlled. For this reason, when an inkjet printer is used as the coating device **500**, a masking process or the like is unnecessary, so that coating by the inkjet printer is preferable. (Adhesive)

[0048] The adhesive or pressure-sensitive adhesive **600** is not particularly limited as long as the adhesive or pressure-sensitive adhesive **600** contains components that do not react with the cavity plate **160** in the vulcanization molding step.

[0049] The coating area of the adhesive or pressure-sensitive adhesive **600** may be determined depending on the use of the base material **140**. For example, when emphasis is placed on adhesion or pressure-sensitive adhesion of the seal member **172**, the coating of the adhesive or pressure-sensitive adhesive **600** may be performed such that the coating area becomes more than or equal to the area of the groove **162** shown in FIG. 3(a). On the other hand, when emphasis is placed on the prevention of contamination of the cavity plate **160** and/or the base material **140**, coating may be performed such that the coating area becomes less than the area of the groove **162**. The same applies to the coating amount of the adhesive or pressure-sensitive adhesive **600**.

[0050] In this manner, the base material **140** is prevented from being contaminated by coating the rubber material **150** with the adhesive or pressure-sensitive adhesive **600** in the coating step, the rubber material **150** being injected into the groove **162** of the cavity plate **160**. In addition, when the adhesive or pressure-sensitive adhesive **600** is dried or in a baking or residue removal step after molding, the contamination of the seal member **172** can be prevented.

[0051] As described above, according to the present embodiment, it is possible to provide the molding method capable of reducing contamination of the base material caused by the adhesive or pressure-sensitive adhesive.

[0052] A preferred embodiment of the present invention has been described above; however, the present invention is not limited thereto, and various modifications or changes can be made without departing from the concept of the present invention.

[0053] In the above-described embodiment, the configuration in which the cavity plate **160**

includes the escape groove **164** and the connecting grooves **166** have been described; however, the cavity plate may not include the escape groove and the connecting groove, and the rubber material **150** may be injected into the groove **162** of the cavity plate in the preforming step.

[0054] In the above-described embodiment, the case where the seal member **172** is molded onto the base material **140** that is flat has been described; however, the shape of the base material **140** may be other shapes. For example, embodiments of the present invention can also be applied to a base material having a flange shape at an end portion, a base material having an uneven shape, or the like. In addition, embodiments of the present invention can also be applied to a case where the seal member **172** is molded onto both surfaces of the base material **140**.

[0055] In addition, for example, embodiments of the present invention includes the following concepts.

Concept 1

[0056] A molding method according to at least one embodiment of the present invention is a molding method for molding a seal member onto a base material, the method including: a first step of injecting a raw material for the seal member into a groove at a temperature where the raw material is not crosslinked, in a first mold including the groove into which the raw material is injected; a second step of coating a surface of the raw material with an adhesive or a pressure-sensitive adhesive, the raw material being injected into the groove of the first mold; and a third step of sandwiching the base material between the first mold on which the surface of the raw material is coated with the adhesive or the pressure-sensitive adhesive and a second mold, and molding the raw material onto the base material as the seal member via the adhesive or the pressure-sensitive adhesive at a temperature where the raw material is crosslinked.

Concept 2

[0057] In the second step, the coating of the adhesive or the pressure-sensitive adhesive may be performed by an inkjet coating device.

Concept 3

[0058] The first mold may include an escape groove in the vicinity of the groove and spaced apart from the groove.

Concept 4

[0059] The first mold may include a connecting groove connecting the groove and the escape groove.

Concept 5

[0060] The base material may be metal, resin, or paper.

Concept 6

[0061] The seal member may be rubber.

[0062] A list of reference numbers follows: [0063] **100**: forming machine [0064] **200, 1000**: molding machine [0065] **110, 112, 210, 212, 1010, 1012**: heating plate [0066] **120, 130, 1020, 1032**: mold [0067] **122, 1022**: pot [0068] **124, 1024**: gate [0069] **133, 1033**: recess [0070] **140, 1040**: base material [0071] **142, 144**: surface [0072] **150, 1050**: rubber material [0073] **160**: cavity plate [0074] **162, 1026**: groove [0075] **164**: escape groove [0076] **166**: connecting groove [0077] **168**: touch surface [0078] **172**: seal member [0079] **174**: side lip [0080] **180**: plate [0081] **222**: second mold [0082] **500**: coating device [0083] **510**: head [0084] **512, 522**: rail [0085] **520**: table [0086] **530**: nozzle [0087] **540**: control unit [0088] **600**: pressure-sensitive adhesive [0089] **610**: bottle

Claims

1. A molding method for molding a seal member onto a base material, the method comprising: a first step of injecting a raw material for the seal member into a groove at a temperature where the raw material is not crosslinked, in a first mold including the groove into which the raw material is

injected; a second step of coating a surface of the raw material with an adhesive or a pressure-sensitive adhesive, the raw material being injected into the groove of the first mold; and a third step of sandwiching the base material between the first mold on which the surface of the raw material is coated with the adhesive or the pressure-sensitive adhesive and a second mold, and molding the raw material onto the base material as the seal member via the adhesive or the pressure-sensitive adhesive at a temperature where the raw material is crosslinked.

2. The molding method according to claim 1, wherein in the second step, the coating of the adhesive or the pressure-sensitive adhesive is performed by an inkjet coating device.
 3. The molding method according to claim 1, wherein the first mold includes an escape groove in the vicinity of the groove and spaced apart from the groove.
 4. The molding method according to claim 1, wherein the first mold includes a connecting groove connecting the groove and the escape groove.
 5. The molding method according to claim 1, wherein the base material is metal, resin, or paper.
 6. The molding method according to claim 1, wherein the seal member is rubber.
 7. The molding method according to claim 2, wherein the first mold includes an escape groove in the vicinity of the groove and spaced apart from the groove.
 8. The molding method according to claim 2, wherein the first mold includes a connecting groove connecting the groove and the escape groove.
 9. The molding method according to claim 3, wherein the first mold includes a connecting groove connecting the groove and the escape groove.
 10. The molding method according to claim 7, wherein the first mold includes a connecting groove connecting the groove and the escape groove.
 11. The molding method according to claim 2, wherein the base material is metal, resin, or paper.
 12. The molding method according to claim 3, wherein the base material is metal, resin, or paper.
 13. The molding method according to claim 4, wherein the base material is metal, resin, or paper.
 14. The molding method according to claim 7, wherein the base material is metal, resin, or paper.
 15. The molding method according to claim 8, wherein the base material is metal, resin, or paper.
 16. The molding method according to claim 2, wherein the seal member is rubber.
 17. The molding method according to claim 3, wherein the seal member is rubber.
 18. The molding method according to claim 4, wherein the seal member is rubber.
 19. The molding method according to claim 5, wherein the seal member is rubber.
 20. The molding method according to claim 7, wherein the seal member is rubber.
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